

Draw a quiver and compute

Click the canvas to add vertices, drag between vertices to add arrows, type relations, pick a field, choose what to compute. Everything is computed exactly by this app's own engine.

Preset

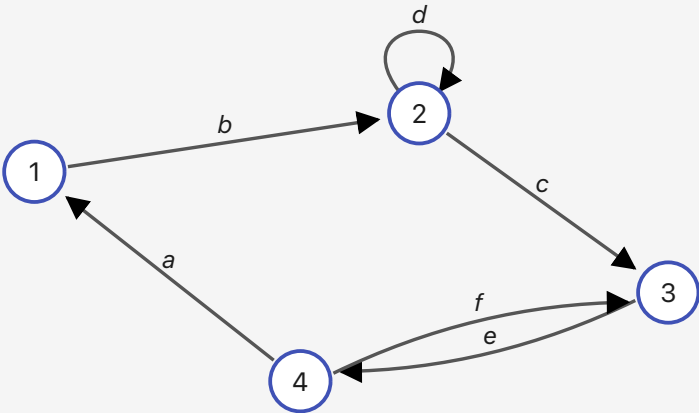
— build your own —

 Field

CC

Clear

engine ready — quiverlab 0.1.0.dev0



Click empty space: add a vertex. Drag vertex → vertex: add an arrow (onto itself: a loop). Click: select. Double-click an arrow label: rename. Delete key: remove.

Relations

e*f, d*d*d,a*b*c,b*d

☒ HH^0..

10

☒ HH_0..

10

☒ Cartan matrix

☒ Coxeter polynomial

☒ gl.dim

☒ center

☒ worked-steps report

☒ Module (no code)

build

I(v) injective

 side

right

injective at vertex

1

☒ dim vector

☒ rad/top/soc

☒ τ

☒ τ^-

☒ proj.dim

☒ inj.dim

☒ proj.res 0..

10

☒ inj.res 0..

10

☒ decompose

☒ Ext 0..

10

☒ Tor 0..

10

second argument N (Ext / Tor target)

build

P(v) projective

side

right

projective at vertex

3

Ext uses N as typed; Tor reads the same N as a LEFT module.

dim = 32

Compute

Cancel

Print report

Report (HTML)

Report data (JSON)

TikZ

JSON

Copy Python

Config (YAML)

Algebra of dimension 32 over CC (computing exactly in QQ)

Hochschild cohomology

n	0	1	2	3	4	5	6	7	8	9	10
dim HH^n	2	4	5	3	3	3	3	3	3	3	3

Chouhy-Solotar
Hochschild, Gerhard (1945). On the cohomology groups of an associative algebra. Annals of Mathematics. Second Series 46, 58-67.

Hochschild homology

n	0	1	2	3	4	5	6	7	8	9	10
dim HH_n	6	2	2	2	2	2	2	2	2	2	2

Chouhy-Solotar
Hochschild, Gerhard (1945). On the cohomology groups of an associative algebra. Annals of Mathematics. Second Series 46, 58-67.

Cartan matrix:

$$C = \begin{pmatrix} 2 & 2 & 1 & 1 \\ 3 & 6 & 3 & 3 \\ 1 & 1 & 1 & 1 \\ 2 & 2 & 1 & 2 \end{pmatrix}$$

Hochschild, Gerhard (1945). On the cohomology groups of an associative algebra. Annals of Mathematics. Second Series 46, 58-67.

$$\chi(t) = t^4 + \frac{4t^3}{3} + 2t^2 + \frac{4t}{3} + 1$$

Hochschild, Gerhard (1945). On the cohomology groups of an associative algebra. Annals of Mathematics. Second Series 46, 58-67.

>= 32 (certified lower bound; not resolved within depth 32)

Hochschild, Gerhard (1945). On the cohomology groups of an associative algebra. Annals of Mathematics. Second Series 46, 58-67.

$$\dim Z(A) = 2$$

Hochschild, Gerhard (1945). On the cohomology groups of an associative algebra. Annals of Mathematics. Second Series 46, 58-67.

radical / top / socle:

	dim vector
rad M	{1: 1, 2: 2, 3: 1, 4: 1}
top M	{1: 1, 2: 1, 3: 0, 4: 1}
soc M	{1: 1, 2: 0, 3: 0, 4: 0}

rad M, arrow a: (1)

rad M, arrow b: $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$

rad M, arrow c: (1 0)

rad M, arrow d: $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

rad M, arrow e: (1)

rad M, arrow f: (0)

top M, arrow a: (0)

top M, arrow b: (0)

top M, arrow d: (0)

soc M: every arrow acts as zero

input M is indecomposable

input M is indecomposable

Green, Edward L.; Solberg, Oyvind; Zacharia, Dan (2001). Minimal projective resolutions. Transactions of the American Mathematical Society 353, 2915-2939. · Assem, Ibrahim; Simson, Daniel; Skowro'nski, Andrzej (2006). Elements of the Representation Theory of Associative Algebras. Volume 1: Techniques of Representation Theory. Cambridge University Press 65.

Krull–Schmidt decomposition — 1 indecomposable summand(s):

summand	multiplicity	dim vector
M_1	1	{1: 2, 2: 3, 3: 1, 4: 2}

Assem, Ibrahim; Simson, Daniel; Skowro'nski, Andrzej (2006). Elements of the Representation Theory of Associative Algebras. Volume 1: Techniques of Representation Theory. Cambridge University Press 65.

projective resolution

term	⊕-decomposition
0	$P_1 \oplus P_2 \oplus P_4$
1	$P_1 \oplus P_2^2 \oplus P_3^3$
2	$P_2^2 \oplus P_3^2$
3	P_2^2
4	P_2^2
5	P_2^2
6	P_2^2
7	P_2^2
8	P_2^2
9	P_2^2
10	P_2^2

pd = ∞ (beyond the probed length)

Green, Edward L.; Solberg, Oyvind; Zacharia, Dan (2001). Minimal projective resolutions. Transactions of the American Mathematical Society 353, 2915-2939.

injective resolution

term	⊕-decomposition
0	I_1
1	0

id = 0

Green, Edward L.; Solberg, Oyvind; Zacharia, Dan (2001). Minimal projective resolutions. Transactions of the American Mathematical Society 353, 2915-2939. · Assem, Ibrahim; Simson, Daniel; Skowro'nski, Andrzej (2006). Elements of the Representation Theory of Associative Algebras. Volume 1: Techniques of Representation Theory. Cambridge University Press 65.

Ext to the target module — dim vector {1: 1, 2: 1, 3: 1, 4: 1}:

n	0	1	2	3	4	5	6	7	8	9	10
dim Ext^n	1	4	4	2	2	2	2	2	2	2	2

Green, Edward L.; Solberg, Oyvind; Zacharia, Dan (2001). Minimal projective resolutions. Transactions of the American Mathematical Society 353, 2915-2939.

Tor with the target left module — dim vector {1: 1, 2: 3, 3: 1, 4: 1}:

n	0	1	2	3	4	5	6	7	8	9	10
dim Tor_n	1	0	0	0	0	0	0	0	0	0	0

Green, Edward L.; Solberg, Oyvind; Zacharia, Dan (2001). Minimal projective resolutions. Transactions of the American Mathematical Society 353, 2915-2939.

Feedback

Local limits: CPU cores 8 · RAM detected 16.0 GiB · worker memory cap 12.7 GiB · 1 worker process(es)